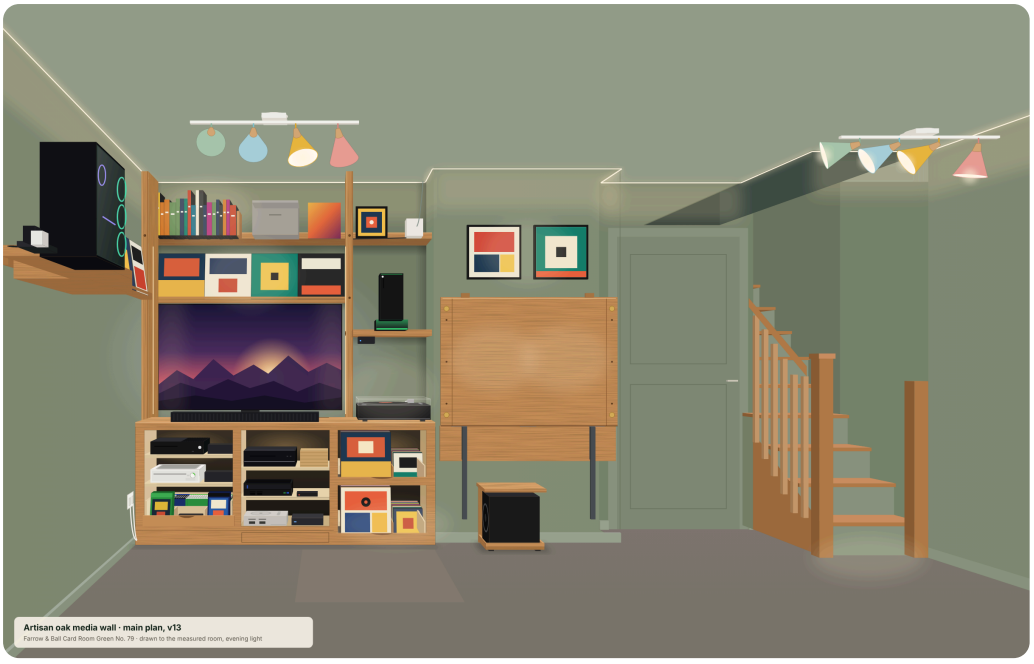
Media Wall & Dining Wall: Build Guide

Version 13 · Oct 18, 2026 · for Emma and Dave · sizes in cm; the full plan has the drawings and the fit proof



This guide assumes you've hung shelves and put flat-pack together, but haven't cut a joint before. Every step says what to do, what to check before moving on, and roughly how long it takes the first time. Where a step is a job for a professional, it says so. Sizes are finished sizes in centimetres.

Before you start: five rules

- **Check every wall before you drill.** Run the cable and pipe detector over the spot, and never drill straight above, below or beside a socket or switch: cables run vertically and horizontally from them.
- **Measure twice, mark once, cut once.** Walls are rarely square: measure the alcove at the floor, 76, 134, 155, 196 and the ceiling, and cut to the smallest.
- **Two people for anything over 15 kg**: the uprights, the worktop, the table top, the TV.
- **Mains wiring is the electrician's.** Swapping the socket, the new socket behind the sofa and the dimmers are all theirs. Adding to existing circuits like this isn't normally notifiable under Part P, but a registered electrician certifies it either way.
- **Dry-fit everything** (assemble without glue or screws) before the final fix.

Tools

You don't need a workshop. Hire the two big tools for a weekend (a track saw and an SDS drill), or ask the timber merchant to cut the oak and ply to the cut lists: most will, for a small charge, and that removes the hardest cuts.

Tool	What it's for	Get it	About £
MEASURE AND MARK			
Tape measure (5 m) and a steel rule	every cut and every height	Buy	10–15
Combination square	squaring lines across boards and checking corners	Buy	10–15

Spirit level, 60 cm and 120 cm	levelling the base, the shelves and the table	Buy	15–30
Cross-line laser level	marking all the heights round the alcove in one go	Buy or borrow	30–60
Cable and pipe detector	checking every wall before you drill: non-negotiable	Buy	15–30
Pencil, masking tape, a marking knife	marking cuts (tape stops oak splintering)	Buy	5
CUT			
Track saw, or a circular saw with a straight-edge guide	the worktop, the table top and the long shelves	Hire or borrow	25–40 a day
Jigsaw	notches and the cable slot	Buy or borrow	40–70
Tenon saw (Japanese pull saw is easiest)	the tenons and the wedge kerfs	Buy	15–25
Sharp chisels, 6, 12 and 25 mm, and a mallet	cleaning out mortises and housings	Buy	30–50
DRILL AND FIX			
Cordless drill-driver (18 V)	pilot holes and screws; most of the job	Buy or borrow	60–120
SDS or hammer drill with masonry bits	plugs into the brick walls and the chimney breast	Hire or borrow	15–25 a day
Dowel jig and 10 mm brad-point bit	the dowels under the uprights and at the wrap ledge	Buy	20–35
Clamps: four 60 cm sash or F-clamps, four small	glue-ups and holding work while you fix it	Buy	40–60
Stud finder (if any wall is plasterboard)	finding studs to fix into	Buy	15
SHAPE AND FINISH			
Router (1/4 in is enough) with a straight bit and a guide	the housings, the LED grooves and the cable groove	Hire, borrow or buy	60–120
Random-orbit sander, 120 and 180 grit	smoothing the oak before oiling	Buy or borrow	40–70
Hardwax oil, lint-free cloths, a brush	finishing the oak	Buy	25–35
SAFETY			
Safety glasses, ear defenders, FFP2 dust masks	every cut, every sanding job; oak dust is an irritant	Buy	15–25
Work gloves; a first-aid kit	handling boards and sheet goods	Buy	10
A sturdy step ladder	the high shelf, the ceiling and the PC shelf	Borrow	—

Six techniques you'll use

1. **Fixing into brick.** Detect, mark, drill with the SDS on hammer setting, clear the dust, push in a brown (7 mm) wall plug and drive a 5 × 60 mm screw. In plasterboard, find a stud, or use heavy-duty plasterboard fixings rated for the load.

2. **Cutting oak cleanly.** Put masking tape over the cut line, score it with a knife, cut with the good face down (circular or track saw) or up (jigsaw). Clamp a straight edge as a guide.
3. **Routing a housing** (a shallow slot a shelf sits in). Clamp a straight batten as a guide, set the bit 4 mm deep, rout, then step down to 12 mm in passes. Practise on an offcut first.
4. **A wedged through-tenon.** Mark the 6 × 2.4 cm mortise on both faces of the upright and drill most of it out with a 20 mm bit from each side. Chisel it to the line. Saw the tenon on the shelf's end. Saw a kerf down its middle, glue it, tap it home, then drive a glued oak wedge into the kerf. When it's dry, saw it flush and plane it.
5. **Dowelling.** Use a dowel jig to drill matching 10 mm holes 3 cm deep in both parts, glue the dowels, clamp.
6. **Oiling.** Sand to 180 grit, wipe off the dust, rub on a thin coat of hardwax oil, wipe off the excess after 15 minutes, let it dry overnight, give it a second coat. Do the undersides too, so the boards don't cup.

The order of work

#	Stage	What to do	Check before moving on	Time
1	**Measure and count**	The alcove's width at the floor, 76, 134, 155 and 196 cm and at the ceiling: cut everything to the smallest. The floor's level across the alcove, and the chimney breast's flatness with a long level. The door's swing, and the sofa: overall width, depth with the return, back height. Count the LPs, singles and games (PS2, Xbox 360, Xbox One), and measure ten cases of each. Make a cardboard template of the alcove at worktop height: walls are rarely square.	Every figure within 1 cm of the plan. If not, send the numbers and the drawings are redone first.	2 h
2	**Network, stage 1**	Unplug everything from the Hub 5. Log in at 192.168.0.1 with the password on its base, then Modem Mode → Enable Modem Mode → Apply. Its page moves to 192.168.100.1. Connect a hub port to Deco A in bedroom 3 and set it up in the Deco app, reusing the old Wi-Fi name and password. If it shows no internet, try another hub port and restart the hub. Plug the new 8-port Switch 1 into Deco A's other port, then the PCs into Switch 1. Add Deco B in the living room and Deco C in the kitchen. For now they link over Wi-Fi. Speed-test each room, and check the Xbox's network settings show NAT: Open.	Internet in every room, and NAT: Open on the Xbox.	2 h
3	**Electrician**	Book one visit for all of it, and get a quote first. Two Cat6 runs from bedroom 3, across the landing and down through the airing cupboard: one to the alcove's right corner for Deco B, one on to the kitchen for Deco C. A spare pulled with each; the optional runs to the master and bedroom 2 if you want them. Swap the single socket on the side wall for a double at the same spot. Add a new double low on the side wall behind the sofa, for the PC and the rear speakers. Chase a 25 mm conduit into the side wall for the PC's leads, from behind the PC shelf to skirting height and behind the skirting to the base, with a draw wire left in it. Make good. LED-compatible dimmers for both ceiling points, both spot bars hung, and a fused spur for the LED drivers. Optional: a conduit to TV height for a future wall mount. Before they go, mark and photograph every cable route on the walls.	The certificate, photos of every route, and both ends of each Cat6 run labelled.	1–2 days (theirs)

4	**Paint**	Paint an A4 board in Card Room Green and look at it by day and with the LED strip on. Wash the skirting and door with sugar soap, sand to a key, and prime them. Two coats of emulsion on the walls, ceiling, alcove and chimney breast. Two coats of eggshell in the same green on the skirting, door, frame and radiator. Leave it a week to cure before fixing anything to it.	Two even coats, and no white showing at the edges or behind the radiator.	2–3 days, then a week to cure
5	**Mark the heights**	With the laser level: 73 and 76 for the worktop, 130 and 134 for the Series X shelf, 153 and 155.4 for the ledges, 192 and 196 for the high shelf. The uprights at 0–4.5 and 131–135.5 cm from the side wall. The dining wall's backboard at 51.3–73.3, centred on the breast. Run the cable detector along every line you will drill, and mark anything it finds.	Every line level end to end, and the uprights' lines plumb.	1 h
6	**Build the base**	Build the 9 cm plinth for bays 1–2, with the open bay for the hidden drawer under bay 2. Level it with shims. Assemble the two BESTÅ frames with their back panels left off and stand them on the plinth. Build bay 3 from 18 mm ply, 60 × 40 × 73, and bolt it to bay 2. Screw the 4 cm spacer blocks to the wall, push the base against them and fit the anti-tip brackets. Lay the flat two-compartment trunking in the channel behind (mains one side, data the other) and run the cables in.	The tops level within 1 mm across 186 cm, and nothing rocks.	1 day
7	**Fit the worktop**	Cut the worktop and the dining wall's backboard from the one 300 × 62 oak board, using the template from step 1. Scribe the ends to the walls; cut the TV slot over the channel and the grommet hole behind the deck. Oil it on every face, undersides and cut ends included. Screw it up through the frame tops in slotted holes, so the oak can move.	Tight to both walls, and the slot sits over the channel.	½ day
8	**Fit out the base**	Shelves on their pins at the fit-proof heights for the consoles. The game drawer in bay 1: runners on the frame's sides, then the box (cut list in the Media wall tab). The record trays in bay 3: runners at 2.3 and 37.0 cm, then each tray. The hidden plinth drawer under bay 2.	Every drawer runs smoothly, and the two trays clear each other by 2 cm.	1–2 days
9	**Uprights and shelves**	On the bench: cut the housings and the mortises in both uprights, and the LED groove and cable groove. Dry-fit each upright with its shelves before any glue. Lay Deco B's leads in the groove behind the right upright. Stand the uprights on their dowels and fix the concealed brackets to the back wall. Fit the high shelf and the ledge, then the Series X shelf: wedge its tenons, and peg the housings.	Uprights plumb, shelves level, and the TV slides in with 2 cm each side.	2–3 days
10	**Side wall and PC shelf**	Screw the oak cleat for the wrap ledge and the PC shelf to the side wall. Dowel the wrap ledge to the left upright's front edge. Fit the concealed PC shelf brackets, then the shelf. Screw the keyboard tray's housing under it and fit its runners. Fit the brush grommet and pull the PC's leads down the chased conduit with the draw wire.	One level line with the ledge over the TV, and the PC shelf holds your weight leaning on it.	1 day
11	**The dining wall**	Screw the backboard to the breast, plugs every 20 cm. Fit the two Hebgo brackets on its face, 80 cm apart, arm tops level with its top. Screw the lift-off hinges to the backboard's top edge and the table's back edge, all the same hand. On the bench: oil the table's underside and set the four brass inserts for the legs. Hang the table and fit the two turn buttons above it. Hang the sub's cover on its two keyhole screws between the arms, and stand the sub on its tray sideways: grille to the east, port to the west.	It folds up and down without binding, both brackets click and lock, and it doesn't move when you lean on it.	1 day

12	**Lighting**	LED strip into each upright's groove, wired to its driver on the fused spur. The ceiling strip through its notches in the uprights; the bias light on the back of the TV. Aim both spot bars (the table in the Paint and light tab says which head goes where). The lamp by the sofa, with a 2700K bulb.	From the sofa, no spot head shows in the screen or shines in your eyes.	½ day
13	**Connect and finish**	TV inputs: HDMI 1 the PC, HDMI 2 the soundbar (eARC), HDMI 3 the Series X, HDMI 4 the HDMI switch (Xbox One, 360, PS3, RetroTINK). The PS2 by component into the RetroTINK and by optical to the soundbar; the PS1 by RGB SCART. The deck to the preamp, the preamp to the soundbar's AUX; set the tracking weight; soundbar in Music mode for vinyl. Network stage 2: plug each Cat6 run into its Deco, check Deco B and C show "Ethernet", and give the Series X and PC high priority. Label both outlets of the double socket and each HDMI switch input; set the sub and rear levels from the sofa.	Everything on the network's "done when" list, and every source plays through the soundbar.	1 day

Roughly 12–15 working days spread over a few weekends, plus the electrician and the paint drying. Steps 6–11 can be split: the base and dining wall first, the oak joinery once you're confident.

Where beginners usually go wrong

- **Cutting the worktop to the tape instead of to a template.** Make a cardboard template of the alcove first; walls are rarely square.
- **Skipping the dry-fit.** Gluing a tenon that doesn't fit is the one mistake you can't undo.
- **Oil on one side only.** The board cups within a week.
- **Loading the trays before the anti-tip brackets go in.** A full tray weighs about 32 kg.
- **Folding the table up with something on it.** Clear it first, every time, and keep both turn buttons turned when it's up.

Shopping list

About £2,590–4,907 plus the electrician, in build order.

Status	Item	Qty	Estimate (£)	Paid (£)
1 NETWORK, NOW				
☐ To do	Put the Virgin Hub 5 into modem mode			
☐ To do	Set up Deco A, Switch 1 and the PCs in bedroom 3			
☐ To do	Add Deco B and Deco C over Wi-Fi for now; speed-test each room and the Xbox			
☐ To buy	Cat6 patch leads, 0.5–2 m	11	15–20	
☐ To buy	TP-Link TL-SG108 8-port switch: one for the living room, one to replace the 5-port in bedroom 3	2	36–50	
2 ELECTRICIAN AND CABLING				
☐ To do	Book one electrician visit: Cat6 from bedroom 3 down through the airing cupboard to the alcove's right corner (Deco B) and on to the kitchen (Deco C), a spare pulled with each, and the optional runs to the master and bedroom 2; swap the side-wall socket for a double **at the same spot**; a new double socket low on the side wall behind the sofa for the PC and the rear speakers **(new)**; LED-compatible dimmers for both ceiling points; a fused spur for the LED drivers; fit both spot bars, on the existing rose between the TV and the sofa and the existing point by the door **(changed: no new point)**; optionally a conduit to TV height for a future wall mount **(from v6.1)**		Quote	
☐ To buy	Cat6 solid pure copper, round (not CCA): two runs, four with the optional ones	2–4 runs	30–60	
☐ To buy	6-way surge-protected extension for the retro set (Xbox One, 360, PS3, PS2, PS1, RetroTINK) **(changed: switched at the wall)**	1	20–35	
☐ To buy	4-port USB charger, 5 V 2 A per port (HDMI switch, fan stand, pad dock, bias light)	1	10–20	
☐ To buy	USB plug, 5 V 2 A or more, for the RetroTINK on the retro 6-way	1	5–10	
☐ To buy	Cable kit: brush grommets, velcro ties, labels	1	20–30	
☐ To buy	Flat two-compartment trunking for the 4 cm channel behind the base, about 2 m	1	10–20	
☐ To do	Same electrician visit: chase a 25 mm conduit into the side wall for the PC's leads, from behind the PC shelf down to skirting height and behind the skirting to the base's plinth; make good and paint (the wall is solid) **(new: replaces the spine and trunking)**		Quote	
☐ To buy	25 mm oval conduit and a draw wire for the chased PC drop, about 4 m; one oak-coloured brush grommet for the PC shelf **(new)**	1	10–25	
☐ To buy	PC tidy kit: braided sleeve, right-angle C13 lead, shelf-notch grommet	1	10–20	
3 PAINT				
☐ To do	Sample Farrow & Ball Card Room Green No. 79 on an A4 board; check it by day and with the LED strip on			
☐ To buy	Card Room Green: Estate (or Modern) Emulsion for walls, ceiling, alcove and chimney breast; Estate Eggshell for skirting, door and frame; primer and undercoat		200–500	
☐ To buy	Woodwork primer, sanding pads and sugar soap for the white skirting and door	1	20–40	
☐ To do	Paint the whole room before anything is fixed: alcove and breast included; skirting and door from white to eggshell			
4 BASE UNIT				
☑ Done	Check the alcove back wall for fixings (solid)			

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To do	Measure the alcove's width at the floor, 76, 134, 155 and 196 and the ceiling, and the floor's level across it			
☑ Done	Check the side wall where the ledge and PC shelf go, and the chimney breast where the dining wall goes: solid (you confirmed) **(new)**			
☐ To buy	Two IKEA BESTÅ 60×40×64 frames, white stained oak effect, for bays 1–2	2	70–200	
☐ To buy	Plinth for bays 1–2: 18 mm ply ladder frame 120 × 38 × 9 cm with a clear bay under bay 2, plus shims	1	15–30	
☐ To buy	Hidden plinth drawer: oak front 54 × 6 with a finger-pull groove underneath, 12 mm ply box 52 × 34 × 5, 300 mm full-extension runners **(changed: no push-to-open at floor level, as v6.1 said)**	1	30–50	
☐ To buy	Bay 3 carcass for the records: 18 mm birch ply, 60 × 40 × 73 cm from the floor, connector bolts **(changed: holds the trays)**	1	45–80	
☐ To buy	Two face-out record tray kits: 12 and 9 mm birch ply cut to size, 6 mm oak dividers, 15 mm oak lips, raked rests, felt, two pairs of 400 mm runners rated 68–70 kg	2	150–240	
☐ To buy	BESTÅ shelves, 56 × 36 cm (20 kg max each), two each for bays 1 and 2	4	12–32	
☐ To buy	Anti-tip brackets and 4 cm spacer blocks: both frames and bay 3	1 set	15–25	
☐ To do	Measure about ten games of each kind (PS2, Xbox 360, Xbox One) before cutting the game drawer			
☐ To buy	Game drawer kit: 12 mm birch ply cut to size, 350 mm full-extension soft-close runners (35–49 kg), oak lip, follower block, felt	1	50–90	
☐ To do	Build the plinth and bay 3, fix everything to the wall, fit the shelves, drawers and trays; place the TV and soundbar			
5 OAK JOINERY				
☐ To buy	Two solid oak uprights, 165 × 30 × 4.5 cm (cut to 159), quarter-sawn if possible	2	150–300	
☐ To buy	Solid oak for the shelves: high shelf in two lengths (128.9 and 51.7 × 30 × 4), Series X shelf 55 × 30 × 4, record ledge 128.9 × 7 × 2.4, wrap ledge 134 × 7 × 2.4, PC shelf 82 × 28 × 4, lips **(changed)**	1 set	200–340	
☐ To buy	Oak wedges, 10 mm oak dowel for pegs and dowels, concealed upright brackets; two concealed shelf brackets for the PC shelf, rated 40 kg+ a pair	1	30–60	
☐ To buy	One solid oak worktop, 300 × 62 cm, full stave: the alcove top (186 × 46) and the dining wall's backboard (110 × 22) from the offcut, chamfered fronts **(changed)**	1	250–450	
☐ To buy	Oak face frame, 20 mm, 186 cm for the base	1	60–120	
☐ To buy	Keyboard tray for the PC shelf: 7 cm oak housing under it, 12 mm ply tray 54 × 26 with an oak lip, 300 mm light-duty runners **(new: back from v6.1)**	1	25–45	
☐ To do	Cut the joints and fit the uprights, shelves, ledges and face frame; notch the uprights for the ceiling strip			
6 DINING WALL				
☐ To buy	Solid oak table top 110 × 80 × 2.7 with pegged breadboard ends, and a 60 × 80 × 2.7 leaf (edge-glued staves from a timber merchant, who can also radius the corners) **(new)**	1 set	220–400	
☐ To buy	Two Häfele Hebgo heavy-duty folding brackets, 287.43.428 (580 arm, 500 kg a pair) **(new)**	2	70–120	
☐ To buy	Union PowerLoad lift-off hinges, 100 × 88, one 3-pack (all the same hand) **(new)**	1	16–20	

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To buy	Eight legs on screw-in plates: tapered oak legs on M8 hanger bolts (to match), or Emuca Ø60 × 710 mm steel legs (cheaper); four for the top, four for the leaf **(new)**	8	70–160	
☐ To buy	Two oak turn buttons, two leaf-alignment dowels and two table clips for the leaf, felt pads **(new)**	1	10–20	
☐ To buy	Four folding chairs, IKEA TERJE or similar (skip if yours fit: seat about 45–48) **(new)**	4	100–160	
☐ To buy	Two LP frames for the chimney breast, hung from 165 cm	2	20–40	
☐ To buy	Four M8 brass threaded inserts for the legs, set into the underside instead of plates (the underside stays plain oiled oak) **(changed: no reeding)**	1	8–15	
☐ To buy	Oak cover for the sub: a 37 × 26.4 × 2 cm top with a drip groove on a 35 × 37.4 × 1.5 backboard, and a 35 × 24 × 2 tray with a 2 cm lip all round; no sides, no front; four rubber feet, two keyhole hangers, hardwax oil **(changed in v13.2: no sides)**	1	30–60	
☐ To do	Fix the backboard to the breast, hang the brackets and the top, check it folds and locks; set the legs to the right height			
7 LIGHTING				
☐ To buy	LED for the upright grooves: 2700K strip, two aluminium channels, driver; 6500K bias light for the TV	1	40–90	
☐ To buy	Two Dunelm Elements Wolston 4-light spotlight bars (£49 each), plus eight warm-white (2700K) dimmable E14 LED bulbs, 6 W max **(changed: two)**	2	120–160	
☐ To buy	A floor or table lamp by the sofa with a 2700K warm LED bulb: the low layer a green room needs **(new: from v6.1)**	1	30–80	
☐ To buy	Later: aluminium channel and opal diffuser for the existing ceiling LED strip (optional)	1	0–80	
8 TV, CONSOLES AND AV				
☐ To buy	5-in-1 4K60 HDMI switch with remote	1	20–35	
☐ To do	Measure the PC-to-TV cable route (about 4–5 m along the skirting)			
☐ To buy	Active optical HDMI cable for the PC, certified Ultra High Speed (48 Gbps), about 5 m	1	20–60	
☐ To buy	High Speed HDMI leads, 1–2 m; two 90° HDMI adapters in case the TV's plugs need them	5 + 2	25–40	
☐ To buy	RetroTINK-6X CE upscaler for the PS2 and PS1 (order on restock; UK price includes VAT and import fees)	1	200–240	
☐ To buy	PS2 component video cable, good quality	1	10–20	
☐ To buy	Optical (TOSLINK) cable, 2 m, PS2 straight to the soundbar	1	5–10	
☐ To buy	RGB SCART cable for the PS1	1	8–15	
☐ To do	Connect the TV inputs; set up the PS2 and RetroTINK			
9 VINYL KIT				
☐ To buy	Replacement stylus for the turntable (exact match for its model)	1	15–40	
☐ To buy	External phono preamp (only if the deck has a PHONO/LINE switch)	1	40–100	
☐ To buy	RCA-to-3.5 mm lead, 2 m: preamp under the Series X shelf to the soundbar's AUX **(changed: shorter)**	1	5–10	
☐ To buy	Isolation feet for the turntable; a carbon-fibre record brush	1	20–40	
☐ To do	Set the tracking weight; soundbar in Music mode			
10 FINISHING				

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To do	Turn on HDMI-CEC (SIMPLINK); PS3 Control for HDMI off			
☐ To do	Deco app: prioritise the Series X and PC, guest Wi-Fi, automatic firmware			
☐ To do	Label both outlets of the double socket (left: everything, right: retro) and each HDMI switch input			
☐ To do	Set the sub and rear speaker levels from the sofa			